

ESTHER PROCTOR

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EDUCATION

Bachelor of Science

Graduated 2024

Mathematics Major

Maryville College

SKILLS/CERTIFICATIONS

Technical Skills

Python, Microsoft Office, LaTeX, Maple, Wolfram Mathematica

Soft Skills

Motivation, Curiosity, Organization, Communication, Problem-Solving, Attention to Detail

Other

TABC Server Permit, Hospitality, Food Service, Math Tutoring

EXPERIENCE

Graduate Teaching Assistant

Aug 2024 - Present

University of Tennessee, Knoxville

Knoxville, TN

- Assist professors in teaching undergraduate mathematics courses, namely, Mathematical Reasoning, Finite Mathematics for Business Majors, and College Algebra (instructor of record).
- Grade homework, classwork, and exams with accuracy and fairness; provide detailed feedback to enhance student understanding.
- Utilize strong communication and organizational skills to manage classroom dynamics and maintain a positive learning environment.

Student Researcher

Mar 2024 - May 2024

Maryville College and Oak Ridge National Laboratory

Oak Ridge, TN and Maryville, TN

- Create a genetic algorithm to prevent malware detection for the purpose of cyber security research.
- Use and develop Python programming skills.
- Attend relevant seminars and network with scientists in the cyber security field.

TartanBand Librarian

Sep 2022 - May 2024

Maryville College

Maryville, TN

- Utilize and develop organizational skills.
- Communicate with players to copy and distribute the correct music.
- File music into its' appropriate location.

Brass Instrument Mute Manufacturer

Oct 2022 - Feb 2024

TrumCor Mutes

Maryville, TN

- Use attention to detail to make mutes for various brass instruments.
- Fulfill orders and input customer data.
- Undergo the process of creating a product from start to finish.

SULI Undergraduate Laboratory Intern

May 2023 - July 2023

Oak Ridge National Laboratory

Oak Ridge, TN

- Studied Gaussian Polynomials in efforts to prove a new combinatorial conjecture.
- Gained exposure to various fields of science and mathematics.
- Attended several weekly seminars and professional development events.

Teaching Assistant

Maryville College

Aug 2022 - Dec 2022 and Aug 2023 - Dec 2023

Maryville, TN

- Assisted students with in-class assignments for Dr. Siopsis' Fall 2022 Pre-Calculus with Trigonometry class.
- Created and delivered Quiz/Practicums and held weekly study sessions for Dr. Smith's Fall 2023 Linear Algebra class.
- Refined my personal knowledge of mathematics.

Private Math Tutor

Private

Aug 2022 - Aug 2023

Maryville, TN; Knoxville, TN

- Tutored middle and high school students in mathematics.
- Used problem solving skills to develop creative ways to help students learn math concepts.
- Grew in my own enjoyment of mathematics as a subject.

AWARDS

Russel N. & Frances Lane Edwards & Troy Bell Lane Waggoner Award

April 15, 2023

Maryville College

Awarded to the most outstanding junior in the department of Mathematics and Computer Science.

William H. Dent Calculus Award

April 15, 2023

Maryville College

Awarded to the student with the most outstanding performance in calculus.

Presidential Scholarship

August 2020 - Present

Maryville College

Mathematics Grades 6-12 Pre-Service Teacher Course Work Scholarship

August 2022 - May 2023

National Council for Teachers of Mathematics

PROJECTS

Studying Unimodality of Over q -Binomial Coefficients This is a research project which I completed with my advisor Dr. Elaine Wong as an intern in the Science Undergraduate Laboratory Internships program at Oak Ridge National Laboratory. This project represents initial progress toward a computer algebra based proof of the unimodality of over q -binomial coefficients. ([View report here](#)) ([View poster here](#)).

Mathematical Model of *Squid Game* Used an absorbing Markov chain to create a mathematical model of the flow of players through the six games from the Netflix show *Squid Game*. My partner and I completed this project using Maple. ([View here](#)).

Mathematical Model of Finances According to College Major Used Excel to model the finances of STEM vs. non-STEM Majors from the beginning of college through the beginning of retirement under various initial conditions. ([View here](#)).

Differential Equations Model of Passenger Pigeon Population Used different forms of the Logistic Population Model (proposed by Pierre F. Verhulst in 1838) to analyze the extinction of passenger pigeons in the early 20th century. I completed this project using Berkeley Madonna. ([View here](#)).